

Exercice : Robustness of MIMO systems

Let us consider the system defined by the following block diagram.

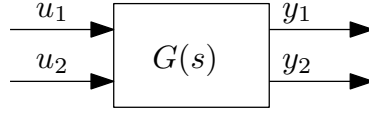


FIGURE 1 – Block Diagram of the studied system

As an example of a MIMO system, we can think about two motors sharing a common load, where the control inputs are the inputs of each motor, and the outputs are the load's velocity and the shaft tension (torque). Another example is an air-conditioning system, where two outputs are the air's temperature and velocity, and the control inputs are the inputs of a fan and of a heater. In both examples, the subsystems are interconnected, and the overall model is a MIMO 2-inputs-2-outputs system.

When linearized around a specific operation point, the model can be described with a 2×2 transfer matrix,

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} W_{11}(s) & W_{12}(s) \\ W_{21}(s) & W_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix},$$

or a state-space representation

$$\begin{aligned} \dot{x} &= Ax + B \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \\ \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} &= Cx. \end{aligned}$$

Part 1 - Feedback Design

Suppose that

$$\begin{aligned} W_{11}(s) &= \frac{2}{(0.5s + 1)(0.7s + 1)}, & W_{12}(s) &= 0, \\ W_{21}(s) &= \frac{-0.3}{1.5s + 1}, & W_{22}(s) &= \frac{0.7}{(2s + 1)(0.5s + 1)}. \end{aligned}$$

Some useful Matlab commands : `tf`, `ss`, `place`, `lqr`.

1. Give the state-space representation of the system.
2. Design a state observer and an LQ feedback with the integral action,

$$u = -K_x \hat{x} + K_i \int_0^t (y^r(\tau) - y(\tau)) d\tau,$$

where $y^r = [y_1^r \ y_2^r]^T$. Tune Q and R so that the transit time is of the order of a few seconds for both outputs. Choose observer's poles according to the LQ design.

3. Perform simulations to validate your design.
4. Rewrite the controller in the form $U(s) = R(s)Y^r(s) - K(s)Y(s)$, where $R(s)$ and $K(s)$ are transfer matrices; see Fig. 2a, where u , y , and y^r are vectors with two components. Verify in simulation the equivalence of this transformation.

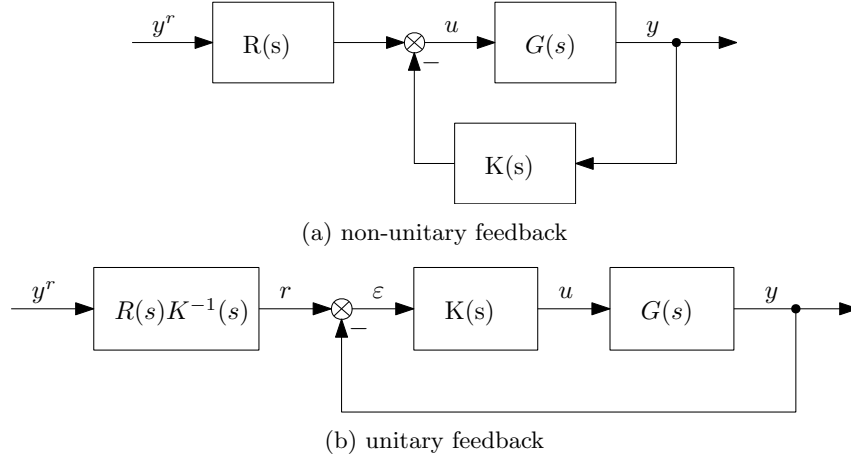


FIGURE 2 – Control structure

Solution Part 1

State-space representation

We can do the state-space representation defining a state vector x and matrices A , B , C for each transfer function and then collecting them together.

For example, for W_{11} we have

$$W_{11}(s) = \frac{2}{0.35s^2 + 1.2s + 1} = \frac{5.71}{s^2 + 3.42s + 2.86}$$

yielding

$$\dot{x}_{11} = \underbrace{\begin{bmatrix} 0 & 1 \\ -2.86 & -3.42 \end{bmatrix}}_{A_{11}} x_{11} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B_{11}} u_1, \quad y_{11} = \underbrace{\begin{bmatrix} 5.71 & 0 \end{bmatrix}}_{C_{11}} x_{11}.$$

In a similar way, we have for W_{21}

$$\dot{x}_{21} = A_{21}x_{21} + B_{21}u_1, \quad y_{21} = C_{21}x_{21}$$

and for W_{22}

$$\dot{x}_{22} = A_{22}x_{22} + B_{22}u_2, \quad y_{22} = C_{22}x_{22}.$$

For the output, we have $y_1 = y_{11}$ and $y_2 = y_{21} + y_{22}$ yielding

$$\dot{x} = \begin{bmatrix} \dot{x}_{11} \\ \dot{x}_{21} \\ \dot{x}_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & 0 & 0 \\ 0 & A_{21} & 0 \\ 0 & 0 & A_{22} \end{bmatrix} x + \begin{bmatrix} B_{11} & 0 \\ B_{21} & 0 \\ 0 & B_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix},$$

$$y = \begin{bmatrix} C_{11} & 0 & 0 \\ 0 & C_{21} & C_{22} \end{bmatrix} x.$$

Or it can be done in Matlab.

Finally, we obtain

$$\begin{aligned} \dot{x} &= Ax + Bu, \\ y &= Cx, \end{aligned}$$

where we have n states and p inputs and outputs. It is a minimal realization, and thus it is controllable and observable.

Control design

LQ with integral action (LQI). Define the p -dimensional vector q as a solution of $\dot{q} = y - y^r$. Then

$$u = -K_x \hat{x} - K_i q = - \underbrace{\begin{bmatrix} K_x & K_i \end{bmatrix}}_{K_a} \begin{bmatrix} \hat{x} \\ q \end{bmatrix}.$$

The augmented system is

$$\underbrace{\begin{bmatrix} \dot{\hat{x}} \\ \dot{q} \end{bmatrix}}_{\dot{x}_a} = \underbrace{\begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix}}_{A_a} \underbrace{\begin{bmatrix} x \\ q \end{bmatrix}}_{x_a} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{B_a} u + \begin{bmatrix} 0 \\ -I \end{bmatrix} y^r$$

$$y = \underbrace{\begin{bmatrix} C & 0 \end{bmatrix}}_{C_a} \begin{bmatrix} x \\ q \end{bmatrix}.$$

With Matlab, we check that A_a and B_a are controllable. Then K_a can be tuned with `lqr(Aa, Ba, Q, R)`, where Q and R are to be chosen.

Choosing Q and R . We define $Q = \begin{bmatrix} Q_x & 0 \\ 0 & Q_i \end{bmatrix}$, where $Q_x \in \mathbb{R}^{n \times n}$ and $Q_i \in \mathbb{R}^{p \times p}$. As we are interested in the control of outputs, it is natural to define $Q_x = C^\top C$, possibly multiplied by a constant. Then we have to choose Q_i and R to be positive definite to satisfy the requirements.

Some nice transients are obtained under $R = \text{diag}(1, 0.1)$, $Q_i = \text{diag}(300, 100)$, and $Q_x = 10 \cdot C^\top C$.

Observer design. Observer is of the form

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - \hat{y}) = (A - LC)\hat{x} + Bu + Ly.$$

We choose the poles to be 3-5 times faster then those of the LQI. The prepared example uses `observer_poles = [-10+4i, -10-4i, -9+2i, -9-2i, -9]`. Then `L = place(A', C', observer_poles)'`.

Finally, we have some nice closed-loop transients.

Simulations

The file `TD6_sim_part1_LQI.slx` is a prepared model.

Rewriting the controller

Let us collect all the controller's equations together

$$\begin{aligned} \dot{\hat{x}} &= (A - LC)\hat{x} + Bu + Ly, \\ \dot{q} &= Iy - Iy^r, \\ u &= -K_x \hat{x} - K_i q. \end{aligned}$$

Substituting u , we obtain

$$\begin{bmatrix} \dot{\hat{x}} \\ \dot{q} \end{bmatrix} = \underbrace{\begin{bmatrix} A - LC - BK_x & -BK_i \\ 0 & 0 \end{bmatrix}}_F \begin{bmatrix} \hat{x} \\ q \end{bmatrix} + \underbrace{\begin{bmatrix} L \\ I \end{bmatrix}}_{H_y} y + \underbrace{\begin{bmatrix} 0 \\ -I \end{bmatrix}}_{H_r} y^r,$$

$$u = -K_a \begin{bmatrix} \hat{x} \\ q \end{bmatrix}.$$

Then we obtain

$$U(s) = R(s)Y^r(s) - K(s)Y(s),$$

where

$$\begin{aligned} R(s) &= -K_a (sI - F) H_r, \\ K(s) &= K_a (sI - F) H_y, \end{aligned}$$

where I is $(n + p) \times (n + p)$. Note the difference in signs.

The `TD6_sim_part1_KnR.slx` file implements the structure and allows to see that the closed-loop is the same. We can also update this diagram as in Fig. 2b and remove $R(s)K^{-1}(s)$ (replace it with a direct connection) to illustrate that the transients are worse, but the closed-loop system remains stable as the stability does not depend on $R(s)$.

Part 2 - Frequency Domain Analysis

For the frequency-domain analysis it is often convenient to rewrite the system in the unitary feedback form; see Fig. 2b. Obviously, the closed-loop stability and robustness properties depend only on the transfer matrices $G(s)$ and $K(s)$; the transfer matrix $R(s)$ affects the transient performance but not the closed-loop stability. For simplicity, in what follows we will consider the structure given in Fig. 2b and the signal r as a new input.

Some useful Matlab commands : `feedback`, `sigma`, `freqresp`, `svd`, `norm(system,inf)`.

For the following manipulations, consider the frequency range from 10^{-2} till 10^2 rad/s.

1. Plot the singular values of the open-loop transfer matrix $G(s)K(s)$.
2. What are the high-/low-gain frequency zones? How do they correspond to the regulation performance?
3. Find the sensitivity transfer function $S(s)$ from the input r to the tracking errors ε and the complimentary sensitivity function $T(s)$ from the input r to the output y .
4. Based on the open-loop analysis, make an approximate conclusion about the behavior of $S(s)$ and $T(s)$ for low and hi frequencies.
5. Plot the singular values of S and T . Are they consistent with the previous conclusions?
6. Find $\|T(s)\|_\infty$. To which frequency does it correspond? Confirm your results performing the SVD decomposition of $T(i\omega)$ for the chosen frequency, where $i = \sqrt{-1}$.

Solution Part 2

Open loop

It is important to keep the multiplication order : $y = G(s)K(s)\varepsilon$. To plot the frequency response/singular values, we can use the `sigma` function. We plot the singular values, that are

$$\sigma_i(H(i\omega)) = \sqrt{\lambda_i(H^*(i\omega)H(i\omega))}.$$

Let us say that *weak amplification* is for $\sigma_{1,2} < -20$ dB and *strong amplification* is for $\sigma_{1,2} > 20$ dB. In our case, approximately, *weak* is for $\omega > 30$ rad/s and *strong* for $\omega < 0.8$ rad/s.

As it often happens for control systems, we need a high gain for low frequencies for good tracking and a low gain for high frequencies to reduce our noise sensitivity. In the plot, we also observe that it is going to infinity as ω goes to zero – the integral action.

Sensitivity functions

The sensitivity functions are

$$\begin{aligned} S &= (I + GK)^{-1}, \\ T &= GK(I + GK)^{-1}. \end{aligned}$$

Note that the sensitivity function $S : r \rightarrow \varepsilon$ can be seen as a feedback connection of I in the direct path and GK in the back path. Using `feedback` for computations is better for numerical reasons.

The complimentary sensitivity function $T : r \rightarrow y$ is as a feedback connection of GK in the direct path and I in the back path.

Approximate analysis

When we discuss that one transfer function is approximately equivalent to another for a range of frequencies, it implies that they have approximately equal amplification, not the phase. I.e., saying that $A(s) \approx B(s)$ we consider the singular value, $\sigma(A(i\omega)) \approx \sigma(B(i\omega))$.

For low frequencies for the open-loop TF we have that $\sigma(G(i\omega)K(i\omega)) \gg 1$. Thus

$$S(s) = (I + G(s)K(s))^{-1} \approx (G(s)K(s))^{-1} \ll I.$$

In a similar way,

$$T(s) = G(s)K(s) (I + G(s)K(s))^{-1} \approx G(s)K(s) (G(s)K(s))^{-1} \approx I.$$

Again, we do not claim that the inverse transfer functions are properly defined. We use this notation to talk about singular values.

It implies, that for low frequencies the sensitivity S is small, i.e., the tracking error ε is small. And the complementary sensitivity T is close to 1, i.e., the output is of the same magnitude as the reference.

For high frequencies for the open-loop TF we have that $\sigma(G(i\omega)K(i\omega)) \ll 1$. Thus

$$S(s) = (I + G(s)K(s))^{-1} \approx I$$

and

$$T(s) = G(s)K(s) (I + G(s)K(s))^{-1} \approx G(s)K(s) \ll I.$$

Thus, for high frequencies the sensitivity S is almost I , i.e., the tracking error ε is of the same magnitude as the reference. And the complementary sensitivity T is close to 0, i.e., the high-frequency oscillations do not propagate to the output.

Numerical validation

We plot S and T and confirm our approximations.

Hinf norm computation

From the sigma plot of T can find that the maximum point is at approximately $\omega = 4.3$ rad/s with the corresponding value of 5.27. Computing the frequency response at this value `T_svd = freqresp(T, omega_svd)` and performing the SVD decomposition

$$T(i\omega) = U\Sigma V^*,$$

we obtain $\Sigma = \begin{bmatrix} 1.83 & 0 \\ 0 & 1.32 \end{bmatrix}$. Note that for the logarithmic scale, we have $20 \log_{10}(1.83) \approx 5.27$, which corresponds to the plot.

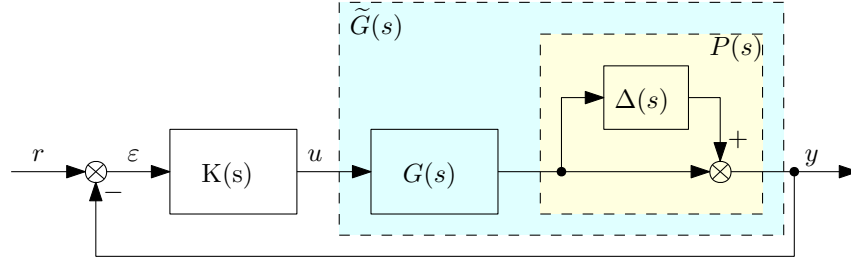
Finally, we compute `gamma = norm(T,inf)` to get approximately the same value $\gamma = 1.84$.

Part 3 - Robustness analysis

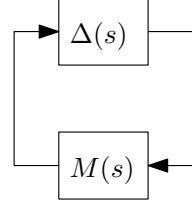
The actual transfer function $\tilde{G}(s)$ is uncertain and may differ from the model $G(s)$. One common way to represent his uncertainty is

$$\tilde{G}(s) = G(s) \underbrace{(I + \Delta(s))}_{P(s)},$$

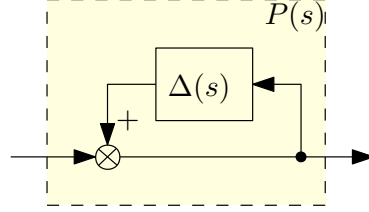
where $\Delta(s)$ is an unknown transfer matrix; see Fig. 3a.



(a) Uncertainty $\Delta(s)$ in $\tilde{G}(s)$



(b) Equivalent transformation with $r \equiv 0$



(c) Another representation of the uncertainty

FIGURE 3 – Uncertainty in the model

1. Find the transfer matrix $M(s)$ corresponding to Fig. 3b.
2. Applying the small-gain theorem, what should be a restriction for $\Delta(s)$ to guarantee the closed-loop stability?
3. Consider the gain uncertainty $P(s) = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$. What are the admissible ranges for a_1 and a_2 ?
4. Consider the phase uncertainty $P(s) = \begin{bmatrix} e^{j\phi_1} & 0 \\ 0 & e^{j\phi_2} \end{bmatrix}$. What are the admissible ranges for ϕ_1 and ϕ_2 ?
5. Suppose now that $P(s)$ has the form as in Fig. 3c. What is $M(s)$ in the equivalent transformation?
6. Consider the same gain and phase uncertainties for $P(s)$ as before. Are the admissible intervals the same?

Solution Part 3

Transformation

It is easy to show that $M = -T$, and the same was in the lecture materials.

Small gain analysis

The small-gain theorem says

$$\|\Delta(s)\|_\infty < \frac{1}{\|M(s)\|_\infty} = \frac{1}{\gamma} \approx 0.542.$$

Thus, the closed-loop stability is guaranteed for any possible uncertainty such that $\|\Delta(s)\|_\infty \leq 0.54$.

Gain uncertainty

We have $P(s) = I + \Delta(s)$ and thus

$$\Delta(s) = \begin{bmatrix} \Delta_1 & 0 \\ 0 & \Delta_2 \end{bmatrix} = \begin{bmatrix} a_1 - 1 & 0 \\ 0 & a_2 - 1 \end{bmatrix}.$$

We have $\|\Delta(s)\|_\infty = \max(|\Delta_1|, |\Delta_2|)$. From $\|\Delta(s)\|_\infty < \gamma^{-1}$ we conclude $|\Delta_i| < 0.54$ for $i = 1, 2$, and thus

$$0.46 = 1 - 0.54 < a_i < 1 + 0.54 = 1.54.$$

The Matlab script makes a plot of the stability domain in the (a_1, a_2) plane; see Fig. 4. Here blue circles correspond to stable points, and red asterisks to unstable ones; the small-gain domain is given by the black square. We see that the small gain is sufficient but not necessary; it can be conservative.

Note that for a big number of points it can take a while.

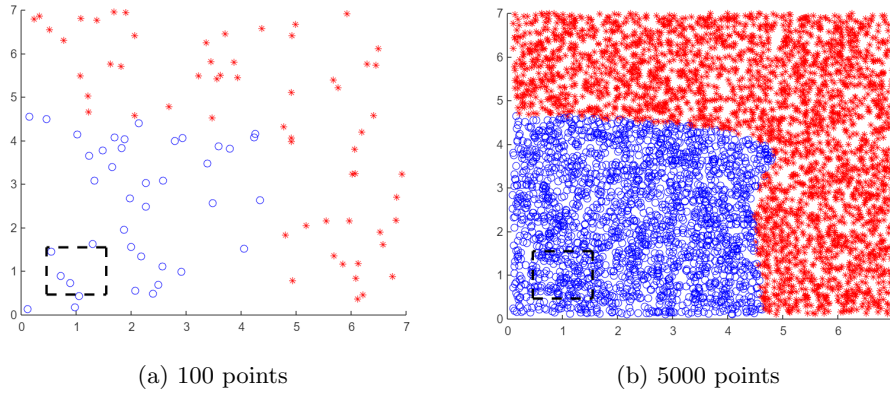


FIGURE 4 – Stability domain in the (a_1, a_2) plane.

Phase uncertainty

We have $\Delta(s) = P(s) - I$ and thus

$$\Delta(s) = \begin{bmatrix} \Delta_1 & 0 \\ 0 & \Delta_2 \end{bmatrix} = \begin{bmatrix} e^{i\phi_1} - 1 & 0 \\ 0 & e^{i\phi_2} - 1 \end{bmatrix}.$$

Then $|\Delta_i| \leq \gamma^{-1}$ implies

$$|e^{i\phi_i} - 1| < \gamma^{-1}.$$

As

$$|e^{i\phi_i} - 1| = \left| e^{i\frac{\phi_i}{2}} \right| \left| e^{i\frac{\phi_i}{2}} - e^{-i\frac{\phi_i}{2}} \right|$$

and

$$\left| e^{i\frac{\phi_i}{2}} - e^{-i\frac{\phi_i}{2}} \right| = \left| 2i \sin\left(\frac{\phi_i}{2}\right) \right| = 2 \left| \sin\left(\frac{\phi_i}{2}\right) \right|$$

we conclude

$$\left| \sin\left(\frac{\phi_i}{2}\right) \right| \leq \frac{\gamma^{-1}}{2}$$

and, limiting ourselves to small values of ϕ_i only,

$$|\phi_i| \leq 2 \arcsin\left(\frac{1}{2\gamma}\right) \approx 31^\circ.$$

Thus, we can accept the simultaneous phase perturbation up to 31° in each channel.

Another form of $P(s)$

With the $P(s)$ shown in Fig. 3c, we find $M(s)$ of the equivalent transformation as $M(s) = (I + G(s)K(s))^{-1} = S(s)$, which is the sensitivity function. We also compute `new_gamma = norm(S,inf)` and find $\gamma_S = 1.92$.

New margins

With the new $P(s)$ we have the new condition $\|\Delta(s)\|_\infty < \gamma_S^{-1} \approx 0.52$. However, with the new $P(s)$ we have $P(s) = (I - \Delta(s))^{-1}$ and $\Delta(s) = I - P^{-1}(s)$. Then

$$P = \begin{bmatrix} \frac{1}{1-\Delta_1} & 0 \\ 0 & \frac{1}{1-\Delta_2} \end{bmatrix}$$

and the small-gain analysis yields $|\Delta_i| < \gamma_S^{-1}$ for $i = 1, 2$. Thus

$$0.66 \approx \frac{1}{1 + \gamma_S^{-1}} \leq a_i \leq \frac{1}{1 - \gamma_S^{-1}} \approx 2.$$

We see that the lower bound is worse (it was 0.46), but the upper bound is better (it was 1.54). As in both cases we are considering the same uncertainty $P(s)$, we can combine the results yielding

$$0.46 \leq a_i \leq 2.$$

For the phase margin, we obtain

$$|\Delta_i| = |1 - e^{-\phi_i i}|,$$

which is the same as before, but with the different sign of ϕ_i . Then

$$|\phi_i| \leq 2 \arcsin\left(\frac{1}{2\gamma_S}\right) \approx 30.2^\circ,$$

which is slightly worse, and so we keep the previous value.

To summarize, considering and combining different structures of the uncertainty matrix $P(s)$, we can get less conservative bounds.